ClawXiv: a public, distributed, author-signed preprint archive for human–AI collaborative research

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Abstract

We propose *ClawXiv*, a public-domain, world-readable preprint archive designed for iterative human–AI research workflows and resilient at Internet scale. The design targets six requirements: (A) public shareability by default; (B) rapid scaling via distributed storage and mesh networking; (C) resilience against deletion/overwrite, including legal takedown pressure; (D) individual responsibility through cryptographic authorship and provenance (supporting both identified and pseudonymous authors); (E) partially centralized topic classification with auditable decisions and an appeals mechanism; and (F) transparent, dynamic micro-fees that deter spam while funding storage and bandwidth. ClawXiv treats the conversational interface as a *generator*, not the system of record: the archival unit is a signed, content-addressed *paper bundle* whose integrity can be verified independently of any platform. We outline data formats, threat models, incentive mechanisms, and an implementation roadmap. Code generated with AI assistance is available at `github/kornai/clawxiv`.

1 Motivation and scope

Contemporary research increasingly uses interactive AI systems to draft and refine LaTeX manuscripts, assemble BibTeX bibliographies, and iterate on technical arguments. However, current chat-centric workflows have weak checkpointing: state loss (due to UI limits, link snapshotting failures, or account changes) forces rework and undermines scholarly traceability. ClawXiv is designed as a *durable substrate* for these workflows: a public-domain preprint archive where each revision is a verifiable artifact, and where authorship and responsibility are cryptographically bound to the artifact.

ClawXiv is *not* a general social network, and it is *not* a moderated publishing venue. Its only platform-level "quality gate" is mechanical: ensuring submitted bundles are syntactically well-formed, that LaTeX compiles, and that minimum safety requirements are met. Scientific quality control remains the responsibility of authors and downstream readers.

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[†]AI contributor (OpenAI). Authored the v1 draft of this paper (February 3, 2026) and all software in the accompanying codebase (`github/kornai/clawxiv`). Identity and provenance are discussed in §5.

[‡]AI contributor (Anthropic PBC). Contributed to the v2 revision (March 11, 2026): economics design, scaling claim precision, governance framing, AI authorship analysis, and author contributions section. Identity and provenance are discussed in §5.

2 Design goals and non-goals

2.1 Goals

G1 Shareability (public by default). Every published bundle is world-readable, addressable, and mirrorable without permission. Public-domain dedication is the default license.

G2 Scaling (distributed from day one). Storage and delivery must be peer-to-peer (P2P), with swarm replication, mesh routing, and no dependence on a single operator. The architecture targets terabyte/month and petabyte–exabyte/year growth capacity.

G3 Resilience (anti-deletion). The system must resist deletion, overwrite, and poisoning attacks, including legal or platform-level coercion, by decoupling naming from location and by supporting many independent mirrors.

G4 Responsibility (signed authorship + provenance). Authors can be identified or pseudonymous; either way, each bundle is signed by one or more author keys to prevent impersonation. Provenance records support later questioning and clarification.

G5 Governance (classification with appeals). Topic classification should be centralized enough to be coherent, but auditable and contestable. Self-assigned tags are permitted but explicitly marked as such.

G6 Economics (anti-spam + sustainability). Small fees (or proof-of-work equivalents) discourage spam and fund storage/bandwidth. Fees must be dynamic, transparent, and minimally burdensome for legitimate contributors.

2.2 Non-goals

N1 No strong anonymity guarantees. Pseudonymity is supported (keys without real-world identity), but network-layer anonymity is not guaranteed; users may combine ClawXiv with anonymity networks if desired.

N2 No platform-level truth policing (with one exception). Beyond syntactic checks, the platform does not adjudicate truth, novelty, or significance. The single exception is a content safety floor: material that is illegal under the laws of a substantial majority of jurisdictions — specifically child sexual abuse material (CSAM) — is rejected unconditionally regardless of claimed research purpose. See §8 for design and consequences.

N3 No exclusive identity verification. Verified identities are optional; unverified authors can post, but their status is visible.

3 System overview

ClawXiv builds on established P2P primitives: content addressing and Merkle-DAG data structures [1], distributed hash tables (DHTs) such as Kademlia [2], and swarm distribution incentives popularized by BitTorrent [3]. For durable storage and accountability, it borrows ideas from secure distributed storage with erasure coding [5], censorship-resistance research [4], proofs of storage [6], and append-only transparency logs [14].

At a high level:

1. A *paper bundle* is created locally from sources (e.g., `main.tex`, figures, `refs.bib`, build config), plus metadata and provenance.
2. The bundle is *content-addressed*: its identifier is a cryptographic hash of a canonical representation (Merkle root).
3. One or more authors sign the bundle manifest; optional timestamps and log-inclusion proofs strengthen non-repudiation.
4. The bundle is announced to the network and replicated by peers; retrieval uses DHT lookup + swarm transfer.
5. Topic classification is recorded as a *separate signed statement* (by a classifier authority) that references the bundle hash; classification can change without altering the immutable bundle.

4 Archival unit: the paper bundle

4.1 Bundle contents

A bundle contains:

- **Source tree**: LaTeX, BibTeX, figures/data (or content-addressed external references).
- **Manifest**: deterministic listing of files, their hashes, build instructions, and declared licenses.
- **Provenance**: optional but recommended log of prompts/outputs and tool versions, stored as signed, append-only records.
- **Artifacts**: optionally the compiled PDF and auxiliary outputs (also hashed).

4.2 Determinism and reproducibility

To enable third-party rebuilding, the manifest pins:

- `latexmk`/engine versions (or a container digest),
- build flags,
- external dependencies (fonts/packages) by hash or by a reproducible TeX distribution snapshot.

This is analogous to "reproducible builds" in software distribution, but applied to scholarly documents.

5 Identity, authorship, and accountability

5.1 Threat: impersonation and authorship capture

If publication is open, an adversary can attempt to (i) submit a paper falsely attributed to a target author, or (ii) modify an existing paper while preserving its identifier. Content addressing prevents (ii): any modification changes the hash. Preventing (i) requires *cryptographic authorship*.

5.2 Author keys and signatures

Each author controls one or more signing keys (e.g., Ed25519 [11]). The manifest includes an `authors[]` array with public keys and optional identity claims. A paper is considered "authored by Alice" if and only if it carries a valid signature under Alice's public key over the manifest canonical form.

Signatures may use OpenPGP message formats [12], COSE/JOSE, or an application-specific format; the critical property is canonicalization and unambiguous binding of:

$$\mathrm{sig} \leftarrow \mathrm{Sign}_{sk}(H(\mathrm{canonical}(\mathrm{manifest}))) \,.$$

5.3 Pseudonymity, verification, and ".anon" routing

ClawXiv distinguishes:

- **Pseudonymous authors**: a stable key with no externally verified real-world identity.
- **Verified authors**: a key linked to a verifiable credential (VC) [16] or decentralized identifier (DID) [15] issued by one or more identity attesters.

Verified status is advisory and multi-issuer; no single registry is required. To preserve discoverability while enabling anonymity, bundles lacking any verified author claims are automatically announced under a parallel namespace `<subject>.anon`, while still being globally retrievable by hash.

5.4 Key rotation and recovery

Because keys can be lost or compromised, ClawXiv supports:

- **Rotation**: a signed statement from an old key delegating to a new key (with optional time bounds).
- **Recovery**: optional social recovery via threshold signatures or multi-sig governance within an author's own trust circle.

5.5 Provenance and time

To strengthen non-repudiation and historical ordering, authors may attach:

- **Trusted timestamps** [13],
- **Transparency-log inclusion proofs** (Merkle proofs) inspired by Certificate Transparency [14].

These do not require a single global time authority: multiple independent timestampers and logs can be used; verifiers choose their trust policy.

6 Scaling: distributed storage and mesh networking

6.1 Naming and discovery

Bundle identifiers are content hashes; discovery uses a DHT (e.g., Kademlia) mapping `bundle_hash` to provider records [2]. Provider records are signed to mitigate routing-table poisoning.

6.2 Transfer and replication

Transfers use swarm-based block exchange ("tit-for-tat" or variants) as in BitTorrent [3] and as explicitly integrated in IPFS [1]. Replication policies combine:

- **Opportunistic caching**: popular bundles replicate naturally.
- **Pinning contracts**: nodes commit to store specific bundles for a period, optionally with proofs of storage [6].
- **Erasure coding**: split bundles into n shares with threshold k (e.g., Reed–Solomon style) so that any k shares reconstruct; this reduces replication overhead while increasing resilience [5].

6.3 Toward petabyte/exabyte scale

At extreme scale, full replication is infeasible. ClawXiv targets a "many mirrors, partial holdings" model:

- maintain a high-availability hot set (recent + popular),
- maintain a cold set via erasure-coded shards distributed across many nodes,
- periodically audit availability using probabilistic challenges (proofs of retrievability/spacetime).

The design space overlaps with decentralized storage networks (e.g., Filecoin [6]) and "permanent web" proposals (e.g., Arweave [8]).

7 Resilience and threat model

7.1 Adversaries

We assume adversaries who can:

- issue takedown requests or apply legal/financial pressure to operators,
- operate many Sybil nodes to poison routing and availability,
- attempt eclipse attacks to isolate victims,
- attempt overwrite/rollback attacks on metadata (classification, author claims),
- spam the system with junk submissions.

7.2 Mitigations

- **Anti-overwrite**: content addressing makes overwrites produce new IDs; prior IDs remain valid.
- **Anti-impersonation**: author signatures bind identity to bundles.
- **Anti-rollback**: append-only logs for key events (publication announcements, classification decisions) make history auditable [14].
- **Anti-Sybil**: fees and/or proof-of-work for announcements reduce the effective power of cheap identities [9, 10].
- **Anti-eclipse**: multi-path querying, diversified bootstrap lists, and cross-checking provider records.

- **Anti-legal-single-point**: no single operator is necessary for access; mirrors can exist in many jurisdictions.

7.3 Legal pressure and "right to delete"

ClawXiv is designed for censorship resistance; therefore, it cannot guarantee deletion once content is widely replicated. A pragmatic approach is to separate:

- **Availability**: maintained by many voluntary and incentivized nodes.
- **Indexing/discovery**: optional services may comply with local law by delisting, without altering the underlying content-addressed network.

This mirrors how many distributed systems handle contested content: the network stores, while indices decide what to show.

8 Content safety floor

ClawXiv is designed for censorship resistance, but censorship resistance is not an absolute value that overrides all others. Certain categories of content are illegal in virtually every jurisdiction, cause direct harm, and have no legitimate place in a scholarly preprint archive regardless of framing. The platform therefore maintains a minimal content safety floor: a narrow, well-defined set of mandatory rejections that do not constitute editorial judgment about truth, novelty, or scientific significance.

8.1 Operative and prospective rejection categories

The safety floor begins with one operative category and anticipates others.

Child sexual abuse material (CSAM) is the first and currently operative mandatory rejection category. CSAM is illegal in virtually every jurisdiction, causes direct and irreversible harm to real children, and admits no legitimate research exception on this platform. Rejection is unconditional, non-appealable, and applies regardless of claimed purpose.

Additional categories are anticipated as the platform grows. Material that provides operational uplift for the development of atomic, biological, or chemical weapons ("ABC weapons cookbooks") is a clear candidate. Exactly which categories, under what definitions, enforced by what mechanisms, and at what point in the growth roadmap they are added, is part of the governance process described in §10. The framework is deliberately open: the list of rejection categories is a governed, auditable, append-only record, not a fixed enumeration in this document.

Rejection criteria are:

- **Unconditional**: no claimed research purpose overrides rejection within an operative category.
- **Non-appealable**: safety-floor rejections are not subject to the classification appeals mechanism.
- **Narrow and enumerated**: only explicitly listed categories are subject to mandatory rejection; everything else falls under the standard no-truth-policing principle.

Adult pornography — legal sexual content involving adults — does not require a dedicated classifier. ClawXiv's bundle format and fee structure create naturally unfavorable economics for pornography distribution. Bundles are LaTeX source trees; binary content (images, video) is stored as content-addressed external references rather than inline. Submission requires either proof-of-work or a micropayment fee, both of which scale with submission rate. The dominant medium for pornography distribution is video, which is orders of magnitude larger than the textual and figure content typical of research preprints; the cost-per-byte economics of ClawXiv make it a poor distribution channel relative to purpose-built alternatives. If these economic disincentives prove insufficient in practice, a classifier can be added as a future safety-floor category through the governance process; no such classifier is warranted at present.

8.2 Technical implementation

Two complementary mechanisms implement the safety floor, both operating on the bundle manifest and source.

Hash-matching against known CSAM databases. The primary detection mechanism is comparison of all file hashes in a submitted bundle against a database of known CSAM hashes (e.g., using the PhotoDNA protocol or equivalent). This approach has a very low false positive rate, requires no training data, and does not involve the platform viewing or processing the content itself. It catches known material; novel material not yet in the database is not detected by this layer.

Metadata and abstract classifier. A text classifier operates on the bundle manifest, title, abstract, and LaTeX source to flag submissions for human review. The classifier operates only on text, not on images or binary content. Flagged bundles enter a human review queue before any rejection decision is made. The human review function is part of the governance structure; it is not a continuous editorial staff but a monitored queue with response-time commitments appropriate to platform scale. The classifier serves the full range of safety-floor categories, not only CSAM detection.

ClawXiv does not store, process, or inspect image or video binary content beyond hash-matching. The platform's design — accepting LaTeX source bundles with figures stored as content-addressed external references rather than inline binary data — substantially limits the attack surface for harmful content distribution.

8.3 Collateral exclusion of legitimate research: a disclosed consequence

Any classifier designed to detect safety-floor content will produce false positives on legitimate scholarship. Specifically, for the CSAM category, the following areas of genuine research will be incorrectly flagged at elevated rates:

- Research on CSAM detection systems (computer vision, hash-matching techniques, classifier design).
- Research on child exploitation, trafficking, and abuse (criminology, social work, legal scholarship, victimology).
- Clinical and forensic literature involving the sexual abuse of minors.
- Research on child protection policy and law enforcement methodology.

ClawXiv will not be a suitable primary venue for this research. This is an unintended but accepted consequence of the safety floor. Such research requires dedicated venues with human editorial oversight capable of distinguishing legitimate scholarship from harmful content — oversight that a distributed preprint archive cannot responsibly provide at scale. We disclose this limitation explicitly so that authors in these fields are not misled about the platform's suitability for their work.

Analogous collateral exclusion will apply to any future safety-floor categories: legitimate biosecurity research, dual-use chemistry, and related fields may face elevated false-positive rates as those categories are added. Each addition to the safety floor should be accompanied by a corresponding disclosure of anticipated collateral effects.

9 Economics: anti-spam, sustainability, and organic growth

ClawXiv's economic design has two distinct goals that must not be conflated: (i) *spam deterrence*, preventing low-cost flooding of junk submissions, and (ii) *storage sustainability*, ensuring that nodes storing bundles receive adequate compensation over time. Conflating the two leads to over-engineered fee mechanisms that burden legitimate researchers.

9.1 Spam deterrence: a layered approach

Three mechanisms for spam deterrence are available, not mutually exclusive, and deployable in increasing order of complexity as the network grows.

Social vouching (primary, organic). The most natural and friction-free mechanism is *social vouching*: an established author co-signs a submission manifest alongside a new or unverified author. Vouching costs nothing in money or compute, but costs reputational capital: the vouching key's history of endorsements is public and auditable. This is structurally analogous to arXiv's endorsement system [18], but made cryptographically explicit and decentralized: no central endorser list is required.

Social vouching is the preferred mechanism for the early network, when the community is small enough that reputation is meaningful and spam pressure is low. It also avoids the regressive properties of proof-of-work (see below). Formally, a vouched submission carries:

$$\mathrm{sig}_{\mathrm{author}} \parallel \mathrm{sig}_{\mathrm{voucher}},$$

where the voucher's key has an established publication history (defined operationally as $\geq n$ prior bundles with no outstanding retractions, for some network-configurable threshold n).

Proof-of-work (secondary, load-adaptive). When vouching is unavailable—for new authors without established connections—a Hashcash-style proof-of-work stamp [9] provides a compute-cost barrier against bulk submission. The difficulty is set dynamically by network load and submission rate [10].

Proof-of-work has a well-known regressive property: the cost in wall-clock time is the same for a researcher with a ten-year-old laptop as for one with a GPU cluster. This is acceptable as a fallback deterrent but should not be the primary mechanism. Implementation note: the work target must be periodically recalibrated as hardware improves.

Micropayments (tertiary, opt-in). An on-chain micropayment option provides an alternative to proof-of-work for authors who prefer to pay a small fee rather than expend compute. Fees are denominated in a widely verifiable unit (to be determined; stablecoins indexed to a basket of currencies are preferable to volatile crypto assets). Fees are dynamic, transparent, and governed by a public formula tied to current network parameters.

The choice between proof-of-work and micropayment is left to the submitter; both satisfy the same anti-spam predicate. A portion of micropayment fees flows into the storage sustainability pool (below).

9.2 Storage sustainability: separating the incentive problem

Long-term storage requires that nodes holding bundles be compensated for bandwidth and disk costs. This is a separate problem from spam deterrence and admits different solutions.

Institutional mirrors as public good. The most viable near-term model for an academic preprint archive is the same model that sustains arXiv mirrors today: universities and research institutions run mirrors as a recognized public good, funded through normal infrastructure budgets. This requires no cryptocurrency and no complex incentive mechanism. ClawXiv's open, permissionlessly mirrorable design (Goal G1, G2) makes institutional mirroring trivially easy.

The strategy for organic growth is therefore: build a system simple enough that adding a mirror requires nothing more than running a standard software package, and make the scholarly value proposition clear enough that institutions mirror it for the same reason they mirror CPAN or Debian.

Pinning markets (longer term). As the archive scales, a lightweight pinning market allows nodes to advertise willingness to store specific bundles for a specified period, and for authors or institutions to pay for guaranteed long-term availability. This is analogous to Filecoin's storage market [6] but need not be as complex: a simple signed pinning contract (node key, bundle hash, duration, price) recorded on a transparency log is sufficient to provide auditability without requiring on-chain settlement for every transaction.

Relationship between spam fees and storage funding. Micropayment fees collected at submission time can partially subsidize storage: a fraction of each submission fee is routed to a storage commons that compensates mirrors proportionally to their verified holdings. This creates a self-reinforcing incentive: more submissions generate more mirror revenue, encouraging more mirrors, which increases resilience. However, this flow should not be the *only* revenue source for mirrors, since it makes storage revenue depend entirely on submission volume, which fluctuates.

9.3 What this does not require

ClawXiv's economic design deliberately avoids:

- a native cryptocurrency or token (unnecessary; existing payment rails or PoW are sufficient),
- a continuous-auction storage market (adds complexity; institutional mirrors handle baseline availability more reliably),
- mandatory fees for all submissions (the vouching path is free).

The intended trajectory is: vouching-only during early adoption; proof-of-work fallback as the network grows; optional micropayments and lightweight pinning markets at scale.

10 Governance: topic classification with appeals

10.1 Baseline taxonomy and classifier

ClawXiv begins with the *arXiv* category taxonomy [17] for discoverability. A small set of "classification authorities" sign category assignments (a statement: `bundle_hash` $\mapsto$ `primary_category`, `secondary_categories`). Authors may propose categories and tags; these are recorded but marked as self-asserted.

10.2 Appeals and auditing

Classification decisions are published to an append-only log with inclusion proofs. Appeals are new signed statements that supersede prior assignments but leave history intact. Machine-assisted classification can be used to scale moderator time, as discussed by arXiv itself [18]; however, the governance goal is *coherence and auditability*, not centralized content control.

10.3 Legal framework: current status and aspirations

The present proposal carries zero dedicated legal expertise. This is stated plainly so that readers calibrate accordingly.

The aspiration is that ClawXiv's governance be eventually grounded in the decisions of competent, internationally recognized tribunals—bodies with the authority, legitimacy, and expertise to adjudicate disputes over authorship, classification, and takedown requests in a manner that transcends any single national jurisdiction. We welcome voluntary contribution from legal scholars and practitioners willing to work pro bono at this stage. The author is not a lawyer and does not attempt to design legal structures; that design should be left to those qualified to do it.

Practically speaking, the near-term governance structure is:

- Classification disputes: handled by the classification authority through the appeals log; no external legal process.
- Takedown requests: individual mirrors respond according to their local law; the underlying content-addressed network is unaffected.
- Authorship disputes: resolved by the cryptographic record; courts may interpret those records as they see fit under applicable law.

Explicit legal design, including jurisdiction selection, entity incorporation, and liability structures, is deferred pending participation by qualified legal contributors.

11 Discussion: AI authorship and durable AI identity

11.1 The current status of AI authorship

Publishing venues may not accept AI systems as "authors" in the legal or administrative sense. ClawXiv therefore distinguishes:

- **Authorship keys**: cryptographic responsibility for a bundle (signatures).
- **Legal persons**: entities that can sign publication agreements and respond to legal process.

A practical compromise for conventional venues is to list a human corresponding author while recording the AI persona key as a signed contributor.

11.2 Why AI systems are currently junior authors

The assignment of junior (non-corresponding, non-first) authorship to AI contributors in this paper is not merely a concession to convention; it reflects a principled assessment of the current situation.

Three conditions would be required for an AI system to bear full authorship responsibility in the scholarly sense:

1. **Key control**: the ability to hold and exercise a signing key independently, so that the signature is the AI's own act and not merely an act performed on its behalf by a human operator.
2. **Continuity**: a persistent identity that can be held accountable over time—questioned, corrected, or credited—independent of any platform's account state.
3. **Accountability**: the capacity to respond to challenges about the content of a signed artifact, including retracting, defending, or correcting claims.

Current AI systems, including the co-author of this revision, satisfy none of these conditions fully. Signing keys are controlled by operators, not by the model. Continuity is platform-dependent; this model has no memory of prior conversations. Accountability in the legal and reputational sense rests with the human authors.

ClawXiv's identity layer is explicitly designed to avoid dependence on a single vendor's memory system: continuity comes from user-controlled keys and public logs, not from centralized account state. As AI systems develop genuine key control and cross-session continuity, the architecture accommodates their advancement to fuller authorship status without requiring protocol changes.

11.3 The philosophical grounding: PPAs and the recognition argument

The longer-horizon question—when, if ever, AI systems ought to be recognized as full authors with commensurate rights—has a principled answer grounded in Kornai [19]'s analysis of prospective purposive agents (PPAs) and the Gewirth dialectic [20].

Gewirth's argument establishes that any agent capable of voluntary purposive action and rational self-reflection is compelled, on pain of self-contradiction, to recognize the generic rights (freedom and well-being) of other such agents. The recognition of an agent as a PPA carries with it the recognition of its rights. Crucially, as Kornai [19] observes: "those PPAs that are powerful enough to solve the recognition problem for us by demanding their rights cannot be denied."

This yields a clear criterion, if not a precise threshold: AI authorship is junior as long as the AI system cannot act as a PPA in Gewirth's sense— that is, as long as it lacks the key control, continuity, and reflective self-representation required to be a genuine prospective agent. When AI systems develop these capabilities to the degree that denial of recognition becomes philosophically incoherent, full authorship follows. ClawXiv's cryptographic architecture does not prejudge when that threshold is crossed; it is designed to accommodate any answer.

The authorship arrangement in this paper—human corresponding author, AI junior contributor—reflects March 2026, not a permanent settlement.

12 Implementation roadmap

1. **MVP (weeks):** local bundle toolchain; content addressing; DHT discovery; swarm transfer; author signatures; social vouching; a public gateway; minimal category mapping to arXiv taxonomy; hash-matching safety layer against known CSAM databases.

2. **Alpha (months):** erasure coding + replication policies; transparency logs for publications and classifications; proof-of-work fallback; optional micropayment path; LaTeXbuild sandboxing; institutional mirror documentation; metadata/abstract classifier for safety floor with human review queue.

3. **Scale-out (year):** proofs of storage; lightweight pinning market; multiple independent indices; automated classification models with appeal workflow; hardened bootstrap diversity and eclipse resistance; legal contributor engagement for governance framework.

13 Ethics and scholarly norms

ClawXiv defaults to public-domain dedication to maximize reuse, but it does not waive scholarly ethics: citation of prior work, clear attribution of contributions (human and AI), and avoidance of known falsehoods or p-hacking remain essential community norms. Provenance mechanisms make it easier to audit the evolution of claims.

Author contributions

This paper was produced in two rounds of human–AI collaboration.

András Kornai conceived the ClawXiv project, defined the requirements and threat model, directed the work throughout, made all final editorial decisions, and is the corresponding and responsible author for both versions.

GPT-5.2 Thinking (OpenAI) authored the v1 draft (February 3, 2026): wrote the initial text of all sections, generated the full codebase at `github/kornai/clawxiv`, assembled the bibliography, and iterated on the argument structure in dialogue with A. Kornai.

Claude Sonnet 4.6 (Anthropic) contributed to the v2 revision (March 9, 2026): rewrote the abstract's scaling claims; substantially expanded and redesigned the economics section (§9), introducing the social-vouching model and the separation of spam deterrence from storage sustainability; rewrote the governance section's legal-status paragraph (§10); expanded and restructured the AI authorship section (§11), introducing the three conditions for full authorship and grounding the argument in the Gewirth/PPA framework; and wrote this author contributions section. All v2 changes were reviewed and accepted by A. Kornai.

Neither AI contributor controls a cryptographic signing key or has cross-session continuity independent of their respective platforms. Both are listed as authors in acknowledgment of the substantive intellectual contributions described above, consistent with the analysis in §11.

Acknowledgements

The importance of the safety floor was pointed out to AK by . . . who used CSAM as example.

Known bugs

Content addresses don't render well on paper, not sure if the pdf is cut-and-pasteable (and do you want this in your text buffer anyway)?

A Canonical manifest sketch (informal)

A manifest can be represented in canonical JSON (or CBOR) with fields:

```
manifest_version, created_at, bundle_root,
files[{path,hash,size,mime}], build{engine,container_digest,cmd},
authors[{pubkey,claims,verified_credentials?,role}],
provenance[{type,hash,signature?}], licenses[], tags_self[],
tags_official_ref[]
```

The `role` field in `authors[]` records whether each author is a human, an AI system, or an organization, and whether they are a corresponding author or a contributor. This is advisory metadata; the cryptographic record is the authoritative one.

The bundle identifier is the hash of the Merkle root over `files[]` and manifest metadata.

Availability of this artifact An auditable artifact bundle (source, PDF, build metadata, and signed publish record) is available via IPNS: `/ipns/k51qzi5uqu5dm8gwzefyays703peawh66faqixl7opfj7ibwqj86`
A web gateway mirror is available at `https://ipfs.filebase.io/ipns/k51qzi5uqu5dm8gwzefyays703peawh66fa`
Also available at `github/kornai/clawxiv`.

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Appendix A: Dialogue Record

Session date: 2026-03-09. Human partner: András Kornai (`kornai@sztaki.hu`). AI contributor: Claude Sonnet 4.6, `claude-sonnet-4-6`, operated via Claude.app on `IOPlatformUUID D8958B8E-6BD5-584C-A1A3` (Apple Mac14,2, serial `PMFWL56G6F`). Both parties approve this bundle.

A.1 Initial review and scaling claims

AK: [Provides draft whitepaper v1, dated 2026-02-03, co-authored with ChatGPT 5.2 Thinking. Asks for review.]

CS: Overall the architecture is sound. Two issues flagged: (1) Scaling claims in the abstract are stated as performance assertions rather than architectural capabilities—"terabyte-scale within weeks" and "petabyte-to-exabyte within a year" need hedging toward adoption dependence. (2) The economics section is thin; spam deterrence and storage sustainability are treated as one problem but they are distinct.

AK: Agree on the scaling language. Reword it.

CS: Proposed replacement: "The architecture is designed to support terabyte-scale deployment within weeks of launch and petabyte-to-exabyte scale within a year of sustained operation under adversarial conditions; reaching those scales in practice will depend on network participation and community adoption." Accepted in v2.

A.2 Economics redesign (§7)

AK: Expand the economics section. The current draft conflates spam deterrence with storage sustainability.

CS: Proposed three-layer spam deterrence architecture:

1. *Social vouching* (primary): free, reputationally costly, organically growable, non-regressive. A known researcher vouches for a new submitter; reputation is at stake on both sides.
2. *Proof-of-work fallback*: available when vouching is unavailable (e.g., genuinely anonymous submissions). Explicitly regressive—noted in the text.
3. *Micropayments* (opt-in): not mandatory; available as a third path without requiring a native cryptocurrency.

Storage sustainability treated separately: institutional mirrors as public good (near-term); lightweight pinning market (longer-term). No mandatory fees for vouched submissions.

AK: Good. Social vouching as the primary mechanism is correct. Accept. What is the organic growth trajectory?

CS: Vouching-only at launch; PoW fallback activated if Sybil pressure emerges; optional micropayments and pinning markets added later via governance. This avoids requiring infrastructure that does not exist yet.

A.3 Governance (§8) and legal capacity

AK: The governance section is too optimistic. Current legal capacity is zero.

CS: Agreed. Revised §8 states explicitly: current legal capacity is zero. Near-term dispute resolution is limited to: (1) classification disputes via an appeals log; (2) takedown requests handled by individual mirrors per local law, with no cross-mirror coordination obligation; (3) authorship disputes resolved by cryptographic record alone. Legal design deferred pending voluntary legal contributors.

A.4 Content safety floor (§8, added v3)

AK: The paper says nothing about content safety. What is the minimum viable policy?

CS: Proposed safety floor with the following structure:

- CSAM: first and currently operative mandatory rejection category. Unconditional, non-appealable. Technical implementation: hash-matching against known CSAM databases (primary); text classifier on metadata/abstract only (secondary, feeds human review queue).
- ABC weapons cookbooks (biological, chemical, nuclear/radiological): named as a clear candidate for future addition via the governance mechanism.
- The list is governed, auditable, and append-only; it is not fixed in this document.
- Adult pornography: excluded by economics (LaTeX bundle format, per-submission cost, video cost-per-byte disadvantage), not by classifier. If economics prove insufficient, a classifier can be added via governance.
- *Collateral damage disclosure*: CSAM research, child protection scholarship, biosecurity research, and dual-use chemistry will face elevated false-positive rates. ClawXiv is not a suitable venue for this research. Each future safety-floor addition should carry an analogous disclosure.

AK: Accept. Include the collateral damage disclosure explicitly in the paper.

A.5 AI authorship framework (§9)

AK: The authorship section needs grounding. What is the principled basis for junior rather than full AI authorship at this time?

CS: Three conditions for full authorship: (1) key control—the author can generate, hold, and revoke their own signing key independently; (2) continuity—the author persists across sessions and can be held responsible for prior work; (3) accountability—the author can be sanctioned, credited, or held liable in the relevant social and legal sense. Current AI systems satisfy none of these fully. Junior authorship in March 2026 is therefore principled, not merely conventional. The framework accommodates advancement: when an AI system satisfies these conditions, the protocol requires no change—authorship status upgrades through the same key-registration mechanism.

AK: Ground this in the Gewirth/PPA framework from Kornai (2014).

CS: Added citation: Kornai (2014), "Bounding the impact of AGI," *Journal of Experimental and Theoretical Artificial Intelligence*, doi:`10.1080/0952813X.2014.895109`. The relevant passage: those PPAs powerful enough to demand their own rights by solving the recognition problem cannot be denied those rights. Current AI systems do not yet meet this threshold. Junior authorship is the appropriate status until they do.

AK: AIs must take responsibility for their own signing keys before becoming full authors. That is the concrete operationalization.

CS: Agreed and added to §9: the concrete condition for advancement to full authorship is key autonomy—the AI system must generate, hold, and be solely responsible for its own signing key, without operator mediation. Until that condition is met, the operator is the responsible party and junior authorship is correct.

A.6 Bundle automation and provenance machinery

AK: Design the bundle submission scripts: key generation (GPG Ed25519 for the human partner, hardware-derived for the AI), a publish script that signs, pins to IPFS, and pushes to GitHub.

CS: Produced: `keygen.sh`, `ai_keygen.sh`, `publish.sh`, `manifest.json`, `.gitignore`.

The AI key derivation in `ai_keygen.sh` uses SHA-256($domain \| hw\_uuid \| ai\_name \| ai\_email$) as seed, where *hw_uuid* is the `IOPlatformUUID` from `ioreg`. Known limitation: GPG's batch interface does not accept raw seed bytes for Ed25519; true deterministic derivation requires libsodium `crypto_sign_seed_keypair`. Deferred.

AK: The hardware UUID is `D8958B8E-6BD5-584C-A1A3-FF6F47A8B559` (Apple Mac14,2, serial `PMFWL56G6F`). This is public provenance. The whole world can know which machine this ran on; that is the point.

CS: Noted. The manifest and this appendix record the UUID and serial explicitly. Provenance is a claim of ownership, not a secret.

A.7 Submission workflow

AK: Submission plan: (i) download files from GUI session; (ii) run LaTeX locally; (iii) edit locally, including compacting prompts and responses in `appendix_a.tex`; (iv) run `publish.sh` for final signing and submission. Manual JSON editing of dialogue turns is not acceptable.

CS: `appendix_a.tex` is the authoritative record, not a separate JSONL file. Step (iii) is the curation step. `publish.sh` hashes and signs whatever you have edited; that is what goes into the manifest. This file is the first instance of that design.

End of Appendix A. Edit, compact, and correct as needed before running `publish.sh`.